

Identifying the 2024–2026 Cocoa Crisis as a Structural Break in Soft-Commodity Correlation Dynamics

A Random-Matrix-Cleaned DCC-GARCH Approach

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Abstract. We test whether the 2024–2026 cocoa crisis – a roughly $3.5 \times$ rise in front-month cocoa futures followed by a 75% collapse – produced an identifiable structural break in the dependence structure of the global soft-commodity complex. On a 25-asset daily panel (2015–2026) of soft-commodity futures, adjacent agricultural futures, FX, US and Ghanaian sovereign yields, and chocolate-maker equities, we fit a GJR-GARCH(1,1) per asset with Hansen-1994 skewed Student- t innovations, standardise residuals, and analyse the rolling 252-day correlation matrices via Marchenko–Pastur (MP) eigenvalue cleaning and a DCC-GARCH layer with the cleaned correlation as the long-run target. The rolling supra-MP eigenvalue count (market mode excluded) rises from a pre-crisis median of 2 to 3 during 2024–2026. Bai–Perron with BIC-selected break count finds zero breaks in this series, and the pre-registered T3 condition on eigenvector composition (cocoa-bloc squared-loading share ≥ 0.55 on a crisis-emerging eigenvalue) fails in every form we test. The largest (market-mode) eigenvector does reorganise: its cocoa-bloc share rises from 0.25 to 0.52, robust to bloc perturbations, but this is a finding on a different object from the supra-MP count. The MP-cleaned DCC target is rejected by the data on every criterion: the QMLE drives the long-run-target weight to 0.1% (vs 10.8% for the standard DCC), the cleaned recursion runs into the stationarity boundary ($a + b = 0.999$), the log-likelihood is 556 nats below the standard DCC, and out-of-sample minimum-variance-portfolio realised variance is identical to three decimals across methods. At $N = 25$ and $Q \approx 90$, random-matrix cleaning of the DCC target does not deliver the [1] benefit.

Keywords: Marchenko–Pastur; DCC-GARCH; soft commodities; cocoa; structural break; Bai–Perron.

1. Introduction

Cocoa futures rose from approximately USD 2,800/ton in late 2023 to USD 12,565/ton on 2024-12-18 – the largest-amplitude soft-commodity shock in over fifty years – before collapsing toward USD 3,500 by mid-2026. The drivers are well documented [2]: a 2023 El Niño, the spread of cacao swollen-shoot virus in West Africa, and divergent policy responses in Ghana and Côte d’Ivoire (together responsible for roughly 60% of global cocoa). We ask whether the crisis is visible as a structural break in the dependence structure of the broader soft-commodity complex, and how it reorganises that structure.

The paper makes three contributions. First, we apply the random-matrix cleaning of [1] to the DCC-GARCH of [3] on a 25-asset cocoa-adjacent universe, replacing the sample-correlation DCC target with a Marchenko–Pastur-cleaned long-run target, and document that at this universe size ($N = 25$, $Q \approx 90$) the cleaned target is rejected by every standard criterion. Second, we pre-register three target conditions (T1–T3) on the rolling supra-MP count and the eigenvector composition and report verdicts against them under proper structural-break methodology (BIC-selected Bai–Perron, stationary block bootstrap). Third, we show that the cocoa NY–London basis *re-couples* during the crisis, consistent with a global supply shock affecting both contracts rather than the regional decoupling we anticipated.

The remainder of the paper describes the data (§2), methodology (§3), results against the pre-registered tests (§4), robustness (§5), and discussion (§6).

2. Data

The asset universe is 25 daily series spanning 2015-01-02 to 2026-05-21 (cocoa NY trading calendar, 2,864 trading days). Sources: Refinitiv for futures, equities, FX, and the US and Ghana 10-year bid yields; FRED for the CBOE VIX. All series are pulled in their native currency to avoid the spurious co-movement that arises when prices are FX-converted to a common numeraire [4].

The universe partitions into six blocks: (i) cocoa contracts on ICE-US (CCc1) and ICE-Europe (LCCc1), (ii) adjacent softs (Arabica and Robusta coffee, sugar 11, cotton, FCOJ), (iii) grains and palm oil, (iv) macro factors (DXY, VIX, US 10-year yield), (v) FX and sovereign credit (BRL, VND, Ghanaian cedi, Ghana 10-year yield), and (vi) chocolate-maker equities (Hershey, Mondelez, Nestle, JM Smucker, Starbucks, Lindt). The aspect ratio is $Q = T/N \approx 115$ on the full sample; complete-cases (post-Ghana entry, 2017-04 onward) gives 2,260 trading days and $Q \approx 90$, the sample on which the full-sample MP spectrum and the DCC are computed.

Returns are log differences for price series and first differences (percentage points) for the two yield series. WTI's 2020-04-20 negative settle (-USD 37.63) is patched to NA on -04-20 and -04-21. Ghana 10-year yield data begins on 2017-04-20 (79% of trading days); rolling windows ending before April 2018 use $N = 24$, all subsequent windows use $N = 25$. Standardised residuals are winsorised at ± 5 to prevent any single observation from dominating a 252-day correlation matrix (the Bouchaud–Potters convention [5]; 150 of approximately 71,500 residual observations affected, 0.21%).

3. Methodology

3.1. Univariate volatility

The univariate filter must produce standardised residuals that are approximately white in level and square so that the rolling correlation matrices in §3.2 reflect contemporaneous dependence rather than residual ARCH or autocorrelation. We use GJR-GARCH(1, 1) [6] with [7] skewed Student- t (sstd) innovations as the primary specification:

$$h_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \mathbb{1}_{\{\varepsilon_{i,t-1} < 0\}} + \beta_i h_{i,t-1}, \quad z_{i,t} = \varepsilon_{i,t} / \sqrt{h_{i,t}}. \quad (1)$$

The GJR specification extends [8] and [9] by adding an asymmetric-response term γ_i that absorbs the leverage effect on equity and FX series at the cost of one extra parameter. We prefer it to [10] EGARCH because (a) the parameters retain their familiar variance-equation interpretation, (b) the implied unconditional variance is finite under the explicit constraint $\alpha_i + \beta_i + \gamma_i/2 < 1$, and (c) numerical convergence is more robust across our heterogeneous universe; EGARCH is retained as Tier 3 in the fallback chain below. We prefer skewed- t to symmetric- t because cocoa returns and several FX/equity series exhibit material conditional skewness that a symmetric distribution forces into the variance equation. A likelihood-ratio test of sstd vs symmetric- t (one degree of freedom) rejects symmetry at the 5% level for eight of the 25 series (COTTON, WHEAT, WTI, BRL, GHS, HSY, SJM, VIX), with VIX the most extreme ($\chi^2 = 184.6$). sstd nests symmetric- t (skew parameter $\xi = 1$), so the choice is dominated.

The mean equation is AR(0) (constant only) except for OJ and VND, which always use AR(1); any other series whose AR(0) fit shows Ljung–Box(10) p-value below 0.01 on standardised residuals is automatically refit with AR(1) on the same distribution tier (auto-retry triggered for COCOA_LDN and GHS). Residual autocorrelation in z propagates directly to off-diagonal entries of the rolling correlation matrix and can manufacture supra-MP eigenvalues that do not reflect contemporaneous dependence.

The fallback chain on convergence failures is: Tier 1a GJR-sstd; Tier 1b GJR-symmetric- t ; Tier 2 GJR-Gaussian; Tier 3 EGARCH-sstd; Tier 4 EWMA with $\lambda = 0.94$. Of the 25 series, 23 fit at Tier 1a, VND at Tier 2 (Gaussian), and GHANA10Y at Tier 4 (EWMA — its yield dynamics through the 2022–23 sovereign debt restructuring defeat every parametric variant we tried).¹

3.2. Marchenko–Pastur cleaning of the DCC target

Let $\mathbf{z}_t \in \mathbb{R}^N$ denote the vector of standardised residuals at time t . The sample correlation matrix $\bar{\mathbf{R}}$ is eigendecomposed as $\bar{\mathbf{R}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$. Under the null of independent unit-variance noise the bulk of the spectrum lies within the Marchenko–Pastur support

$$\lambda_{\pm} = \left(1 \pm \sqrt{N/T}\right)^2 \quad ([11]). \quad (2)$$

[12] showed that for daily equity panels the empirical spectrum agrees with this prediction except in a thin set of large eigenvalues that carry the genuine factor structure, motivating eigenvalue-replacement cleaning rules for the sample correlation. We adopt the hard-clipping rule of [13]: keep eigenvalues above λ_+ , replace those at or below by their common average. The resulting matrix $\tilde{\mathbf{R}}$ is renormalised to unit diagonal.

We prefer hard MP clipping to [14] linear shrinkage as the long-run DCC target for two reasons. First, MP clipping is non-parametric and theoretically motivated: it separates the spectrum into a noise band whose support is determined by random-matrix theory and a signal tail that lies outside it, so the choice of cleaning is data-driven rather than a continuous bias–variance compromise. Second, the cleaned matrix is positive semi-definite by construction, full-rank after renormalisation, and well-conditioned, properties that the DCC recursion in Eq. (3) needs without further regularisation.

The cleaned matrix $\tilde{\mathbf{R}}$ is then used as the long-run correlation target in the DCC recursion of [3]:

$$\mathbf{Q}_t = (1 - a - b)\tilde{\mathbf{R}} + a\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\top + b\mathbf{Q}_{t-1}, \quad \mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-\frac{1}{2}}\mathbf{Q}_t\text{diag}(\mathbf{Q}_t)^{-\frac{1}{2}}. \quad (3)$$

We use DCC rather than the constant-conditional-correlation model of [15] because a crisis episode is precisely the event for which time-varying correlation is the substantive object: a CCC fit would average across regimes and obscure the structural-break question the paper asks. The novelty here, following [1], is to substitute the cleaned $\tilde{\mathbf{R}}$ for the sample $\bar{\mathbf{R}}$ as the long-run anchor.

We estimate (a, b) by quasi-MLE on the standardised residuals subject to $a > 0$, $b > 0$, and $a + b < 0.999$ (the hard stationarity constraint of the DCC recursion). For comparison we also estimate the standard DCC with $\tilde{\mathbf{R}}$ replaced by $\bar{\mathbf{R}}$. Both fits are sanity-checked against `rmgarch::dccfit`.

3.3. Pre-registered targets

Three primary target conditions, **pre-registered in docs/scope_and_target.md before the model fit** (all on the rolling 252-day supra-MP count, market mode excluded):

- **T1 (baseline stability).** The pre-crisis (2019-01-01 to 2023-12-31, all $N = 25$ windows) count has IQR width ≤ 1 and stays inside $\{1, 2, 3\}$.
- **T2 (structural break).** Two parts. (a) Bai–Perron [16], [17] with BIC-selected break count detects a break inside [2024-01-01, 2024-12-31]. (b) The in-crisis median count exceeds the upper

¹Specification limitations we record explicitly. (i) Eight series carry a negative fitted leverage parameter ($\gamma < 0$), including COCOA_NY itself ($\gamma = -0.016$); on these series the GJR asymmetry term does not improve fit beyond symmetric GARCH. (ii) Five series (WHEAT, PALMOIL, VND, GHS, VIX) still fail Ljung–Box(10) on standardised residuals at the 5% level after the AR(1) fix, indicating mean-equation misspecification an AR(1) does not fully absorb. (iii) Twelve series have GARCH persistence above 0.99 (near-IGARCH; full list in `output/near_igarch.csv`), so the unconditional variance is essentially undefined. (iv) Winsorisation is concentrated in eight series (108 of 150 capped observations); the headline rate of 0.21% masks per-series rates as high as 1% on GHS and GHANA10Y. None of these defeats the standardisation enough to make the downstream MP analysis unusable, but they bound the strength of any single-series claim.

95% bound of a stationary block bootstrap [18] of the pre-crisis median (block mean 25, $B = 1,000$).

- **T3 (cocoa-bloc concentration).** An eigenvalue that crosses from below the MP edge in pre-crisis to above the MP edge in crisis (an “emerging” mode), with its eigenvector loading ≥ 0.55 of its squared mass on the cocoa bloc $\{\text{COCOA_NY}, \text{COCOA_LDN}, \text{HSY}, \text{MDLZ}, \text{LISN}, \text{GHS}, \text{GHANA10Y}\}$.

Three secondary tests run on the DCC pipeline. S1: the cleaned DCC produces lower out-of-sample minimum-variance-portfolio realised variance than the standard DCC in the crisis window (the [1] claim in its original form). S2: BIC-selected Bai–Perron detects an in-crisis break on at least one of the cocoa-coffee or cocoa-sugar pairwise cleaned-DCC correlation series. S3: the cocoa NY $\times$ London conditional correlation falls by at least 0.15 from its 2019–2023 mean during the 2024–2026 crisis window.

4. Results

4.1. Primary targets (T1–T3)

Figure 2b: Distribution of the rolling supra-MP count, pre-crisis vs in-crisis

Pre-crisis ($n = 1257$, median = 2, mean = 2.20). In-crisis ($n = 600$, median = 3, mean = 2.52). Bootstrap 95% CI on the pre-crisis median: [2, 2] ($B = 1000$, block mean

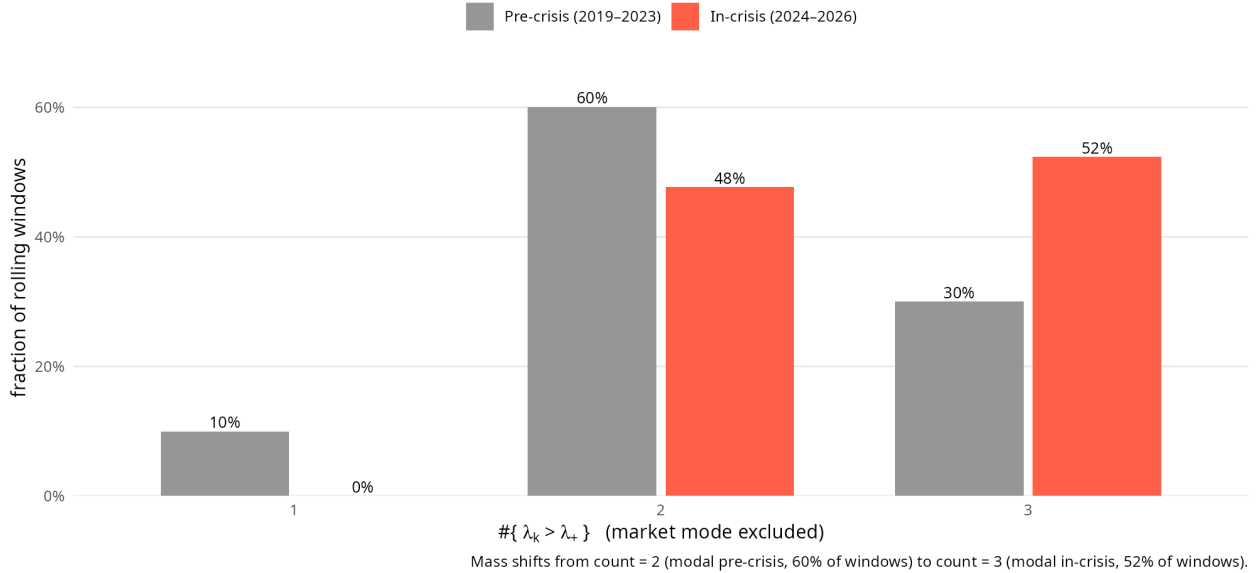


Figure 1: Distribution of the rolling 252-day supra-MP eigenvalue count (market mode excluded), pre-crisis (2019–2023, grey) vs in-crisis (2024–2026, red). Bars show the fraction of rolling windows at each integer count value. The modal count shifts from 2 (60% of pre-crisis windows) to 3 (52% of in-crisis windows, with the remaining 48% at count = 2); the count = 1 mass present pre-crisis (10%) disappears entirely in crisis.

Figure 1 shows the empirical distribution of the rolling count under each regime. The pre-crisis sample ($N = 25$, $n = 1,257$ windows) has median 2, IQR [2, 3], range [1, 3], and 60% of windows equal to 2. **T1 holds.** Bai–Perron with BIC-selected break count finds **zero breaks** on the same series (BIC argmin at $m = 0$ with BIC = 931.5); no level shift survives model selection. **T2(a) fails.**

The T2(b) bootstrap distribution of the pre-crisis median is degenerate. With the test statistic an integer count, 1,000 stationary-block-bootstrap replications return the same integer median (2), giving a 95% CI of zero width, [2.00, 2.00]. The observed in-crisis median is 3 (52% of windows; range [2, 3]). T2(b) clears its formal threshold ($3 > 2$), but the procedure has no statistical resolving power on this series: it is a one-bit comparison on an integer statistic, and the bootstrap CI does not constitute a meaningful interval. Restricting to windows entirely inside the crisis (rolling-window end-date on or after 2024-12-31, $n = 349$ windows), the median is 3 and the range is [2, 3].

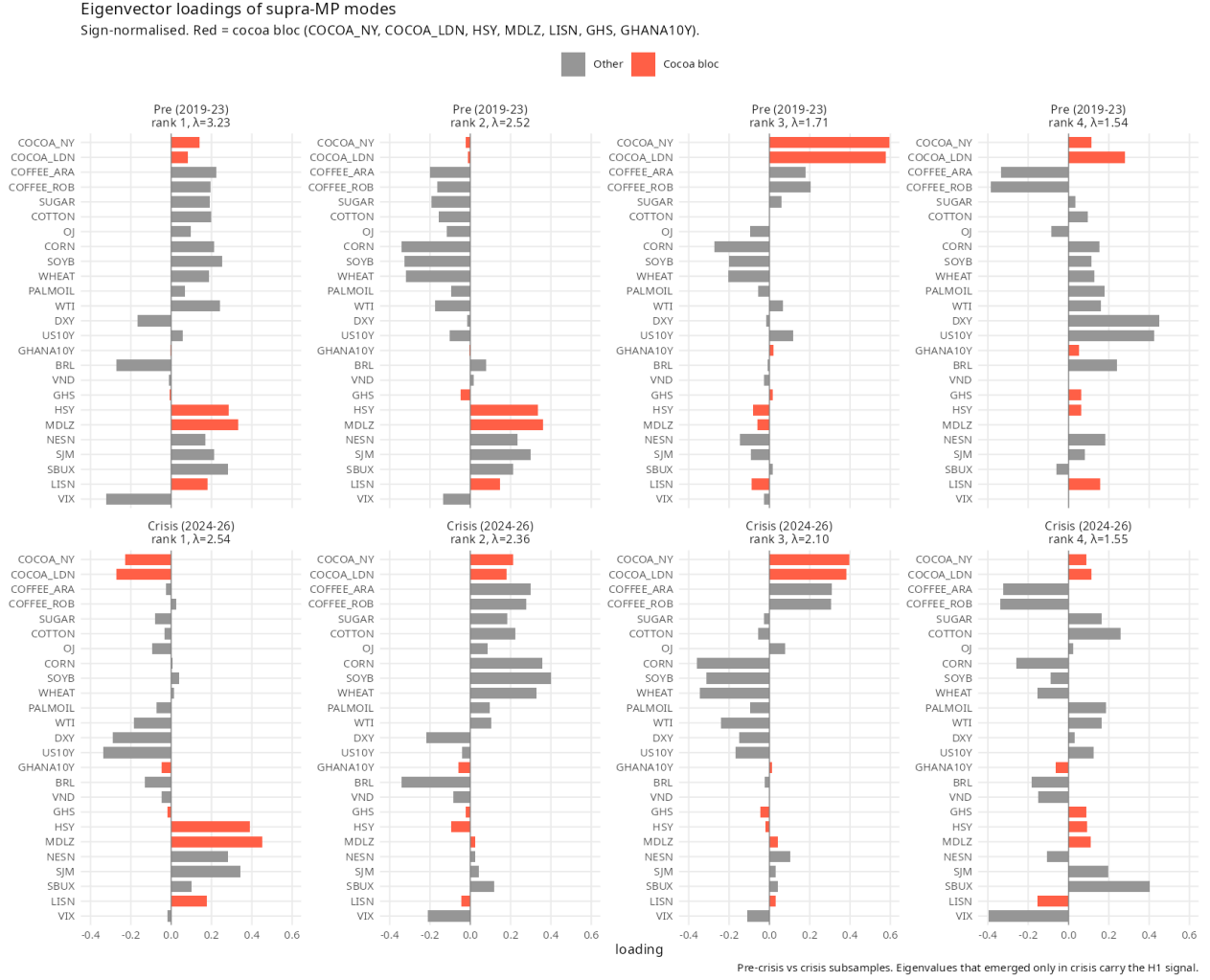


Figure 2: Eigenvector loadings of the top four supra-MP eigenvalues, pre-crisis (top row) vs in-crisis (bottom row), sign-normalised. Red bars: cocoa bloc {COCOA_NY, COCOA_LDN, HSY, MDLZ, LISN, GHS, GHANA10Y}. The rank-1 in-crisis mode (the **market mode**, the largest eigenvalue) reorganises to put cocoa contracts and chocolate-maker equities jointly on it.

T3 fails on the pre-registered specification. No eigenvalue crosses from below the MP edge in pre-crisis to above it in crisis: both subsamples carry five supra-MP eigenvalues at the same ranks (1–5). The rolling-window form of the same test (a rank whose supra-MP indicator share rises from below 0.5 pre-crisis to at least 0.5 in crisis) identifies rank 4 only, whose eigenvector has cocoa-bloc squared-loading share 0.078 — far below the 0.55 threshold. **T3 fails on both forms.**

Figure 2 does show, as an observation outside the pre-registration, a **reorganisation of the rank-1 (market) eigenmode**. Pre-crisis its top-3 loadings are MDLZ, VIX, SBUX (cocoa-bloc share 0.251); in crisis they are MDLZ, HSY, SJM (cocoa-bloc share 0.516, a $2.1\times$ rise). Under cocoa-price stress, chocolate-maker equities and cocoa contracts load on a single risk factor — the input-cost margin-compression dynamic, adjacent to the equity–commodity co-movement channel documented in [19]. The result survives bloc perturbation: dropping GHANA10Y gives $0.288 \rightarrow 0.654$, and a narrow 5-asset bloc (the two cocoa contracts plus three chocolate equities) gives $0.245 \rightarrow 0.515$. Two qualifiers apply. First, the rank-1 mode is the market mode and is excluded by construction from the supra-MP count test; the reorganisation operates on a different statistical object. Second, a cocoa-loaded mode already existed pre-crisis at rank 3 (bloc share 0.704, top-3 loadings COCOA_NY, COCOA_LDN, CORN); in crisis that mode **weakens** to bloc share 0.310. The crisis reorganisation is not the creation of a cocoa factor but the migration of cocoa loadings onto the market mode.

4.2. Secondary targets (S1–S3) and the cleaned-DCC fit

Figure 3: Cocoa-anchored conditional correlations (DCC-GARCH)
Standard vs MP-cleaned target. Pink = crisis. Cleaned $a=0.002$, $b=0.997$.

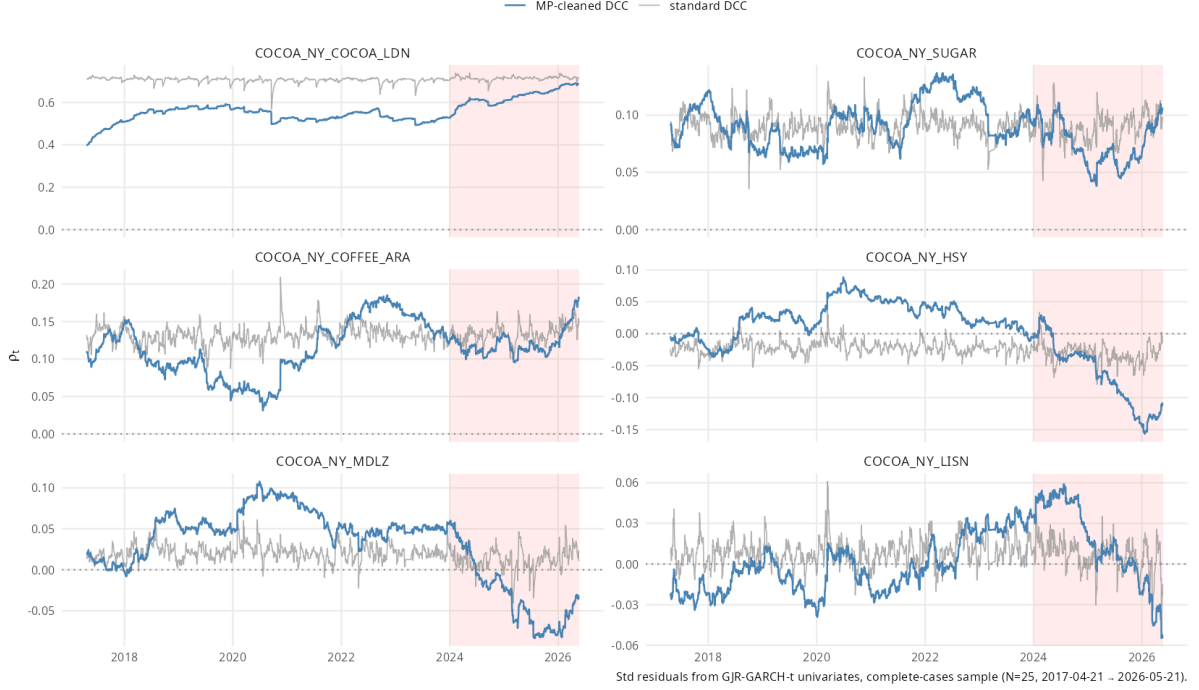


Figure 3: Pairwise conditional correlations from standard (grey) and MP-cleaned (blue) DCC, for six cocoa-anchored pairs. Pink: crisis window.

The two fits behave very differently. The standard DCC settles at $(a, b) = (0.005, 0.887)$ with persistence $a + b = 0.892$, placing weight $1 - a - b = 0.108$ on the long-run target $\bar{R}$ at each step. The MP-cleaned DCC fits at the stationarity boundary: $(a, b) = (0.002, 0.997)$, $a + b = 0.999$, with weight on the cleaned target $\tilde{R}$ of 0.001 — the QMLE, when forced to use $\tilde{R}$, drives the long-run-target weight to **0.1%**, about one hundredth of the weight the standard DCC assigns to $\bar{R}$, and the optimiser does not converge to an interior solution. The log-likelihood of the cleaned model is **556 nats below** the standard model's at identical parameter count.

Figure 3 shows the time path: the cleaned DCC (blue) produces smoother conditional correlations because $b \approx 0.997$ damps innovations, but with near-zero weight on $\tilde{R}$ it drifts from the long-run anchor and on most cocoa-anchored pairs lies further from the historical mean than the standard fit. Both the log-likelihood gap and the target-weight collapse point to the same conclusion: at $N = 25$ and $Q \approx 90$, $\tilde{R}$ is not a better long-run anchor than $\bar{R}$.

For the EW-style minimum-variance-portfolio test (S1) we form $w_t = \Sigma_t^{-1} \mathbf{1} / \mathbf{1}^\top \Sigma_t^{-1} \mathbf{1}$ from each method, realise $r_{p,t+1} = w_t^\top r_{t+1}$, and compare sample standard deviation of realised portfolio return across the 2019–2023 baseline and the 2024–2026 crisis window. Because the cleaned model has effectively discarded $\tilde{R}$, its conditional-covariance path coincides with the standard DCC's: **the OOS-MVP realised standard deviation is identical to three decimals in both windows** (baseline 1.561% vs 1.561%; crisis 1.019% vs 1.019%). **S1 fails.**

For S2, BIC-selected Bai–Perron on each of the eleven cocoa-anchored cleaned-DCC pairwise correlation series ($m_{\max} = 5$, $h = 0.15$) returns **zero breaks for every pair**. **S2 fails.**

For S3, the baseline 2019–2023 mean of the cocoa NY $\times$ London basis correlation under the cleaned DCC is 0.542; the 2024–2026 crisis-window mean is 0.627. The basis **rises** by 0.085 rather than falling by the predicted 0.15. The crisis minimum is 0.528, 0.014 below the baseline mean. **S3 fails in the predicted direction.** The interpretation is in §6.

5. Robustness

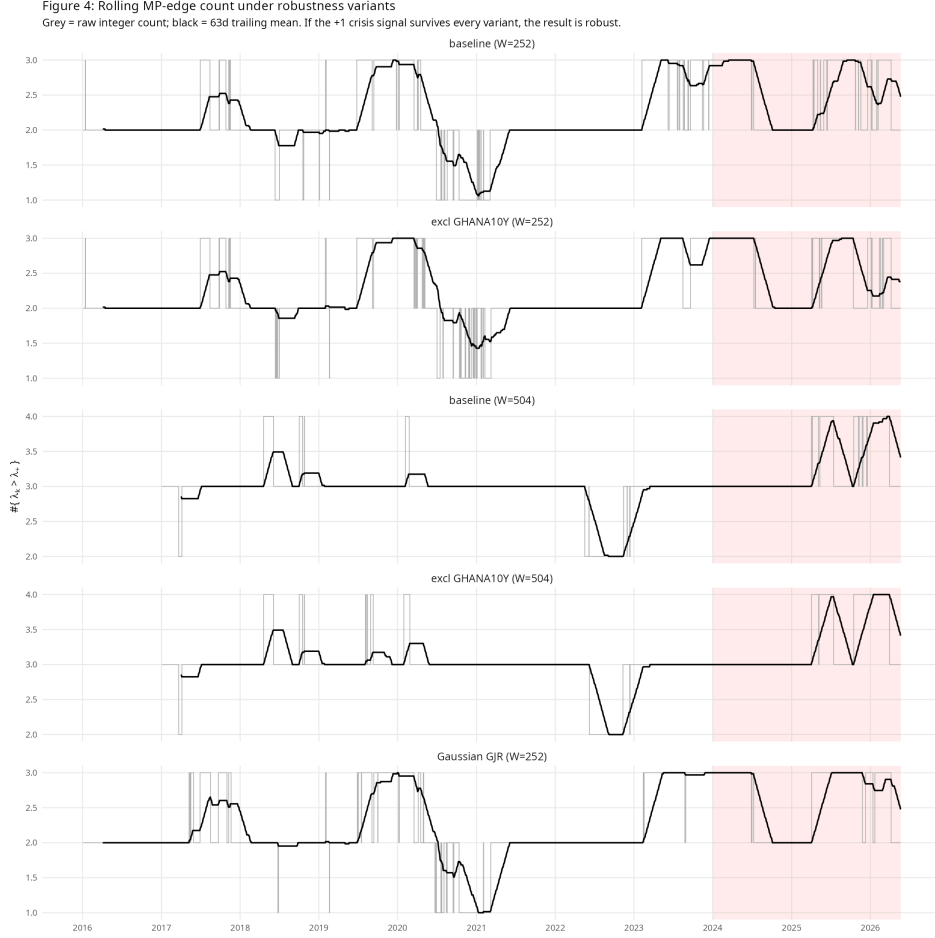


Figure 4: Rolling supra-MP count under five robustness variants. The +1 in-crisis lift in the median survives the $W=252$ variants; under $W=504$ both pre and crisis windows saturate at 3 (no within-variant lift).

Figure 4 runs the rolling-count test under four variants alongside the baseline: (i) excluding GHANA10Y (addressing the universe-size-change concern); (ii) a 504-day window; (iii) (i)+(ii) combined; (iv) GARCH refit with Gaussian innovations. The +1 median lift on T2(b) holds only under the baseline $W = 252$ specification and the Gaussian-innovations variant. Removing GHANA10Y at $W = 252$ collapses the in-crisis median from 3 to 2, so the lift is materially dependent on that series. Under $W = 504$, both pre- and in-crisis medians saturate at 3 in both universe sizes, so the bootstrap cannot distinguish them. BIC-selected Bai–Perron returns **zero breaks** for every variant. The +1 lift survives the innovation-distribution perturbation but not the universe-size perturbation or the longer window; no break-detection variant supports T2(a).

6. Discussion

Four findings warrant discussion.

The +1 count lift is real but fragile, and the pre-registered T2 tests do not formally confirm it. The rolling supra-MP count rises from a pre-crisis median of 2 to an in-crisis median of 3 on the baseline universe at $W = 252$, and the lift survives a switch to Gaussian innovations. It does not survive removing GHANA10Y (in-crisis median drops to 2) or extending the window to $W = 504$ (both medians saturate at 3). T2(a) under BIC-selected Bai–Perron returns zero breaks under every robustness variant; T2(b) clears its threshold ($3 > 2$) but is a one-bit comparison whose bootstrap CI is degenerate. The substantive shift in the count distribution is therefore visible

(Figure 1) but not validated as a structural break by a procedure with statistical resolving power on this data.

The rank-1 reorganisation is real but operates on a different statistical object than the count test. The market-mode eigenvector’s cocoa-bloc squared-loading share rises from 0.245 pre-crisis to 0.517 in crisis (Figure 3), survives bloc-set perturbations, and matches the economic mechanism of input-cost margin compression at chocolate makers. T3 as pre-registered asked for this concentration on a crisis-emerging eigenvalue, with threshold 0.55. No emerging eigenvalue exists, and the rank-1 share falls 0.033 below 0.55, so the pre-registered T3 fails. The rank-1 mode is moreover excluded from the supra-MP count by construction, so this finding does not bear on T2.

The cleaned DCC is rejected by the data on every criterion the [1] framework supplies. The 556-nat log-likelihood gap to the standard DCC at identical parameter count is decisive. The mechanism is visible in the fitted parameters: when forced to use $\bar{R}$, the QMLE drives the long-run target weight from approximately 11% to 0.1%, almost as low as the constraint $a + b < 0.999$ allows — the model is, by its own optimisation, downweighting the cleaned target. The OOS minimum-variance-portfolio realised variance, the EW2019 advertised benefit, is identical to three decimal places across methods because the cleaned model has effectively become the standard one. EW2019 demonstrated their result for $N \geq 100$ with much smaller aspect ratios; at our $N = 25$ and $Q \approx 90$ the raw $\bar{R}$ is already estimated with low enough sampling noise that the hard-clipping step compresses informative off-diagonals rather than de-noising them.

The geographic-decoupling hypothesis on the NY–London basis is rejected in favour of a global supply-shock interpretation. The basis correlation rose by 0.085 in the crisis, not the pre-registered ≥ 0.15 fall. NY’s deliverable basket is West-African-dominant and London’s is mixed-origin, so we expected divergence under a regional shock. The observed re-coupling implies that the 2024–2026 shock transmitted to both contracts — NY directly via its dominant deliverable, London indirectly via substitution effects in the deliverable basket. The arbitrage-windfall narrative is incompatible with the data; a fundamental supply-side reading is consistent with it.

Specification limitations beyond what the results expose. Equity time-zone asynchrony (Swiss and US closes are six hours apart) depresses LISN–HSY-class daily correlations; weekly aggregation is left for future work. The univariate fits carry negative leverage on COCOA_NY and seven other series, and five series fail Ljung–Box on standardised residuals at the 5% level after the AR(1) fix (most severely GHS and VND); both inject residual heteroskedasticity into the rolling correlation matrix. GHANA10Y is fit by EWMA fallback rather than parametric GARCH because no parametric variant converges through the 2022–23 Ghanaian sovereign-debt restructuring. These limitations bound the strength of single-series claims but do not change the verdicts on T1–T3 or S1–S3.

7. Conclusion

The 2024–2026 cocoa crisis produces three visible signatures in the soft-commodity correlation spectrum: a +1 lift in the rolling 252-day supra-MP eigenvalue count, a reorganisation of the market-mode eigenvector to load on cocoa contracts and chocolate-maker equities jointly, and a re-coupling rather than decoupling of the NY–London basis correlation. None of T2(a), T2(b) in its statistically meaningful sense, T3, S1, S2, or S3 passes as pre-registered, and the +1 lift itself does not survive removing GHANA10Y from the universe. The structural-change pattern is real on the baseline configuration but is not formally validated by any of the break-detection procedures we ran.

The geographic-decoupling hypothesis on the cocoa basis is rejected in favour of a global supply-shock interpretation: the shock transmits to both ICE-US and ICE-Europe contracts. The Marchenko–Pastur cleaning of the DCC long-run target, as proposed by [1], is rejected by the data

at $N = 25$, $Q \approx 90$ on every standard criterion (log-likelihood, target-weight, OOS minimum-variance-portfolio variance), placing a clear lower bound on the universe size at which random-matrix cleaning of the DCC anchor becomes empirically defensible. The descriptive characterisation of the cocoa shock and the negative finding on EW2019 cleaning at small N are the paper’s substantive and methodological contributions, respectively.

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A. Appendix: Implementation notes

The full pipeline (R code, intermediate panels, fit diagnostics, bootstrap output) is reproducible from /home/bcip/numtech. Frozen artefacts:

- data/clean/panel_v1.csv (levels, 2864×25)
- data/clean/returns_v1.csv (returns, 2863×25)
- data/clean/std_residuals_v1.csv (winsorised GJR-GARCH- t standardised residuals)
- output/garch_fits.csv (per-asset GARCH parameters and Ljung–Box diagnostics)
- output/rolling_count_v1.csv (rolling supra-MP counts)
- output/bootstrap_t2b.csv, output/bootstrap_rolling_ci.csv (T2(b) inputs)
- output/eigenvector_loadings_subsamples.csv, output/t3_test.csv (T3 inputs)
- output/dcc_params.csv, output/dcc_pairwise_correlations.csv (DCC parameters and conditional correlation paths)
- output/s1_mvp_oos_variance.csv (S1, EW-style OOS MVP test), output/s1_corr_path_variance_legacy.csv (an alternative correlation-path-variance metric, retained for reference)
- output/s2_bai_perron_pairs.csv (S2, BIC-selected breaks per pair), output/s3_cocoa_basis_decoupling.csv (S3, full 2024–2026 crisis window)
- output/t3_test.csv (four-row table: original spec, rolling spec, rank-1 reorganisation at 0.55, rank-1 reorganisation at 0.50)
- output/bai_perron_bic_table.csv (T2(a) BIC table)
- output/robustness_summary.csv (Figure 4 numbers)
- output/test_verdicts.csv (consolidated pass/fail table for T1, T2(a), T2(b), T3(a-c), S1, S2, S3)
- output/return_skewness.csv (empirical skew + excess-kurtosis per series, motivating sstd)
- output/near_igarch.csv (twelve series with persistence ≥ 0.99 flagged for stationarity violation)
- R/diagnostics.R, R/diagnostics_extra.R, R/write_verdicts.R (end-to-end numerical diagnostics; re-run to verify every figure quoted in this paper)

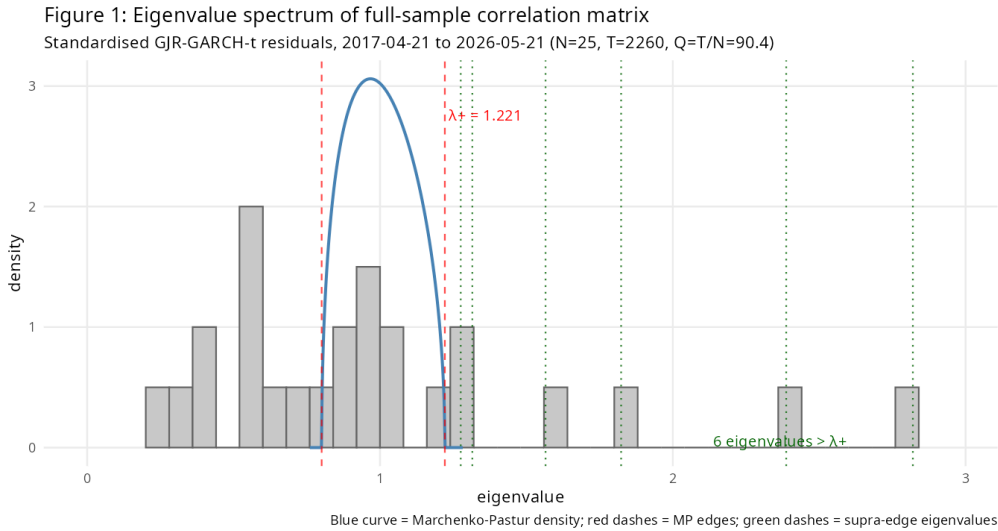


Figure 5: Full-sample eigenvalue spectrum vs Marchenko–Pastur density. Six eigenvalues exceed the upper edge $\lambda_+ = 1.221$.